



IASTAM 6.0

IEEE Industry Applications Society Tunisia Annual Meeting

TECHNICAL CHALLENGE SPECIFICATION BOOK

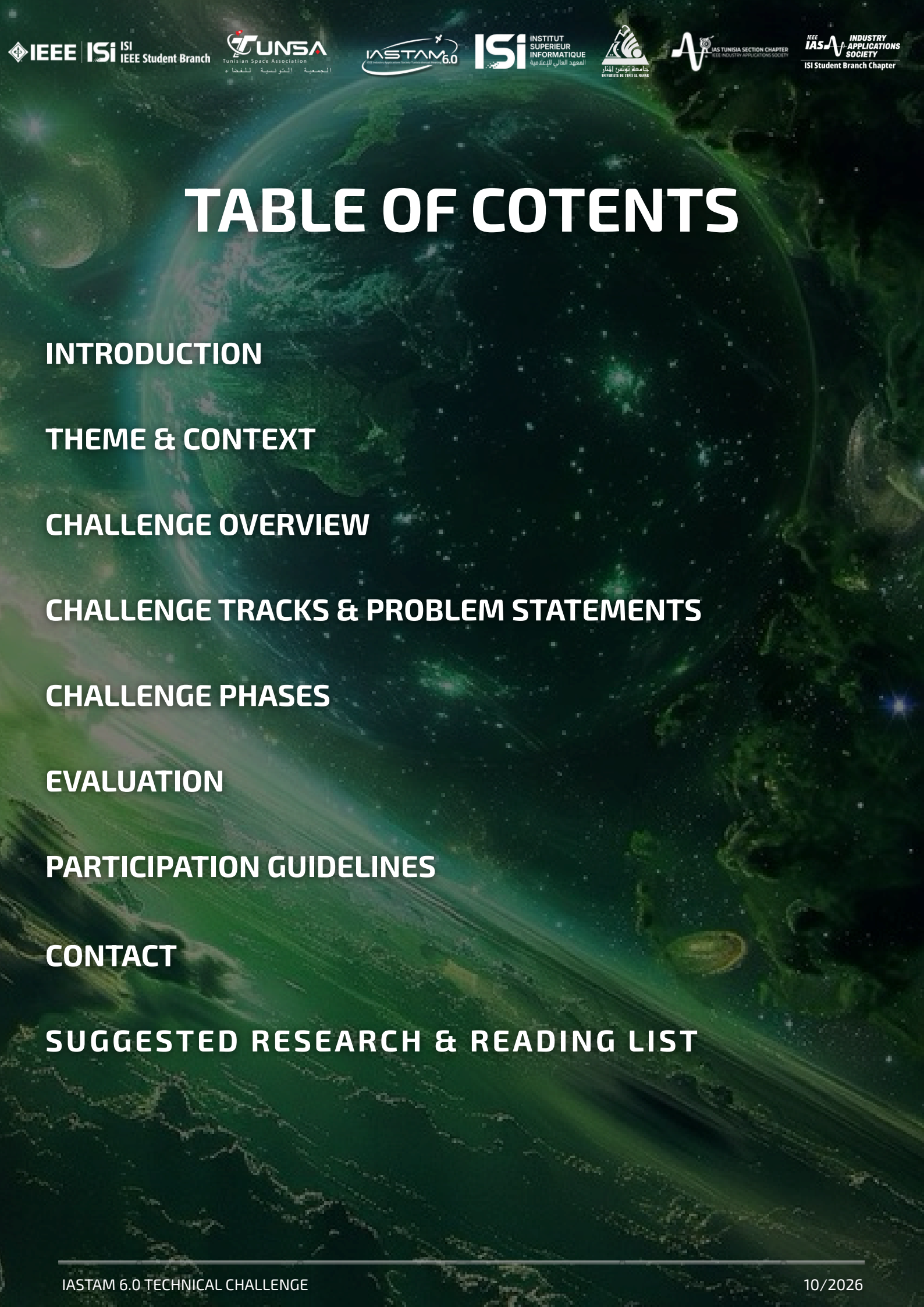


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01. INTRODUCTION

The **IASTAM 6.0** Technical Challenge is a seven-week research-based engineering competition in collaboration with **TUNSA (Tunisian Space Association)**, ending with a final pitch to a professional jury.

Teams tackle a real, unsolved problem by researching, designing, and defending a solution with evidence. The challenge is open to beginners, with clear scenarios, constraints, and measurable performance criteria.

02. THEME AND CONTEXT

2.1. BUILDING THE FUTURE BEYOND EARTH

The 2026 edition focuses on the future of human activity beyond Earth. The goal is shifting from exploration to creation: developing space infrastructure, in-space production, and sustainable off-Earth ecosystems that could transform life and work both in space and on Earth.

2.2. THE PROBLEM: DATA CENTERS BEYOND EARTH

The 2026 edition focuses on the future of human activity beyond Earth. The goal is shifting from exploration to creation: developing space infrastructure, in-space production, and sustainable off-Earth ecosystems that could transform life and work both in space and on Earth.

The world's demand for computing is outgrowing the planet's ability to power and cool it. The race to train and run large AI models has triggered an unprecedented surge in data-center construction, and these facilities are extraordinarily hungry for electricity, water, and land.

This has pushed the industry to look beyond Earth. Orbit offers two things the ground cannot: near-continuous sunlight for power, and the cold of deep space as a natural heat sink. The vision is computing that runs on abundant solar energy and sheds its heat by radiation, without burning through water or grid capacity.

This is no longer hypothetical. In late 2025, Starcloud launched a data-center-class GPU into orbit and became the first to train a language model in space. Google unveiled Project Suncatcher, a plan for constellations of solar-powered satellites carrying AI accelerators linked by laser.

The question is no longer whether compute will move to orbit, but how?

03. CHALLENGE OVERVIEW

2.1. OBJECTIVES

- Enable teams to investigate a real engineering problem and produce a defensible technical solution.
- Develop research skills: literature review, methodology, evaluation, and technical writing.
- Build solutions that are realistic and measurable rather than purely conceptual.

2.2. SCOPE

The challenge is open to national and international participants. Teams choose one problem statement from one of five tracks: artificial intelligence and onboard computing, distributed and federated learning, orbital computing architecture, networks and communications, and cybersecurity.

No prior experience in space systems is required. Each problem is framed so that a team with a solid background in software, machine learning, networking, systems, or security can approach it directly. Teams are supported by mentors throughout the development phase.

2.3. CHALLENGE DESIGN PRINCIPLES

Every problem statement in this book was written to satisfy the following principles. Teams should expect these elements in each challenge, and should structure their own work around them.

- A clearly defined problem.
- A realistic but accessible space scenario.
- Explicit technical constraints.
- Shared data or a common simulation environment where possible.
- Measurable performance indicators.
- A prototype, a model, or a demonstrator as the final deliverable.

04. CHALLENGE TRACKS AND PROBLEM STATEMENTS

The challenge is divided into five tracks, each containing two problem statements. Each team selects one problem statement and works on it for the full duration of the challenge.

TRACK 1: ARTIFICIAL INTELLIGENCE AND ONBOARD COMPUTING

Problem 1: Process or Transmit?

- **Context:** A satellite continuously produces data from its instruments. Its computing resources, energy, memory, and communication opportunities with the ground are limited.
- **Question:** How can the system decide automatically, for each piece of data or task, whether it is preferable to process it immediately onboard, store it for later processing, or transmit it to the ground?
- **Objective:** Develop a decision engine or a scheduling algorithm that optimises the scientific or operational value obtained as a function of the available resources.
- **Possible constraints:** Available energy, CPU/GPU load, memory, storage, data priority, communication windows, bandwidth, and transmission delay.
- **Evaluation:** Quality of the decisions, rate of completed tasks, estimated energy consumption, latency, and resource utilisation.

Problem 2: AI Inference Under Energy Constraints

- **Context:** An AI task must be executed onboard a satellite despite limited resources.
- **Question:** How can an AI inference be executed while respecting severe constraints on computation, energy, and memory?
- **Objective:** Design a strategy that dynamically adapts the model, the inference frequency, or the processing quality to the available resources.
- **Possible constraints:** Computing power, memory, energy, and processing deadlines.
- **Evaluation:** Accuracy, inference time, estimated consumption, memory used, and robustness to variations in available resources.

04. CHALLENGE TRACKS AND PROBLEM STATEMENTS

TRACK 2: DISTRIBUTED AND FEDERATED LEARNING

Problem 3: Learning Despite Connection Losses

- **Context:** A constellation collects data locally. The satellites do not have a permanent connection and some links become temporarily unavailable.
- **Question:** How can a shared model be trained when participants appear, disappear, and communicate intermittently?
- **Objective:** Develop a federated or distributed learning strategy capable of maintaining training progress despite interruptions and data heterogeneity.
- **Possible constraints:** Intermittent connections, unavailable satellites, non-homogeneous data, limited bandwidth and energy.
- **Evaluation:** Final accuracy, convergence speed, volume of data exchanged, robustness to node loss, and estimated energy cost.

Problem 4: Learning With Minimum Communication

- **Context:** Exchanges between satellites are costly and limited.
- **Question:** How can a high-performing global model be obtained while reducing exchanges between satellites as much as possible?
- **Objective:** Propose a method of compression, update selection, or intelligent aggregation that limits the total communication volume.
- **Possible constraints:** Bandwidth, energy, and intermittent connections.
- **Evaluation:** Model performance relative to the volume transmitted, number of exchanges, convergence time, and robustness.

04. CHALLENGE TRACKS AND PROBLEM STATEMENTS

TRACK 3: ORBITAL COMPUTING ARCHITECTURE

Problem 5: Where Should the Computation Take Place?

- **Context:** A mission may have several potential locations to execute a processing task: onboard the satellite, in an orbital infrastructure, in a ground station, or in a terrestrial cloud.
- **Question:** How can the best location be chosen dynamically for each workload?
- **Objective:** Develop a decision engine capable of assigning each task to the most relevant computing location according to the context.
- **Possible constraints:** Latency, energy, computing capacity, network availability, task priority, data volume, and cost.
- **Evaluation:** Total execution time, latency, estimated consumption, resource utilisation rate, and relevance of the decisions.

Problem 6: Dynamic Workload Allocation Between Space and Ground

- **Context:** Computing resources and communication conditions evolve over the course of the mission.
- **Question:** How can a set of evolving tasks be distributed across several resources located in space and on the ground?
- **Objective:** Design an allocation strategy capable of adapting to changing resources and communication conditions.
- **Possible constraints:** Availability, capacity, latency, and workload evolution.
- **Evaluation:** Processing time, load balance, resilience to unavailability, and overall efficiency.

04. CHALLENGE TRACKS AND PROBLEM STATEMENTS

TRACK 4: NETWORKS AND COMMUNICATIONS

Problem 7: Transmitting Information Rather Than Raw Data

- **Context:** A satellite can generate a large volume of images or data while only a small part is genuinely useful to the mission.
- **Question:** How can the volume to be transmitted be strongly reduced while preserving the information necessary for decision-making on the ground?
- **Objective:** Develop a processing chain capable of analysing the data, identifying the relevant information, and transmitting only what carries sufficient value.
- **Possible constraints:** For example: a mission is searching for ships in a given area. Should all images be transmitted, or only the detections and zones of interest?
- **Evaluation:** Data reduction rate, preservation of useful information, accuracy of the results, and latency.

Problem 8: Building a Resilient Satellite Network

- **Context:** The topology of a satellite network evolves permanently and some links may disappear temporarily.
- **Question:** How can efficient data circulation be maintained despite the evolving network?
- **Objective:** Develop or simulate a routing and adaptation strategy that preserves communication despite disruptions.
- **Possible constraints:** Intermittent links, dynamic topology, and limited network resources.
- **Evaluation:** Delivery rate, latency, network utilisation, recovery after disruption, and robustness.

04. CHALLENGE TRACKS AND PROBLEM STATEMENTS

TRACK 5: CYBERSECURITY

Problem 9: Can the Satellite Still Be Trusted?

- **Context:** A space system receives commands, produces telemetry, and executes software remotely. Anomalies must be detected without necessarily interrupting the mission.
- **Question:** How can potentially malicious or abnormal behaviour be detected automatically in a space system?
- **Objective:** Develop an anomaly detection mechanism capable of identifying unusual commands, falsified data, or behaviour incompatible with expected operation.
- **Possible constraints:** Commands, telemetry, network activity, and software behaviour.
- **Evaluation:** Detection rate, false positives, detection time, robustness, and impact on availability.

Problem 10: Detecting an Attack Before Mission Failure

- **Context:** A sequence of weak events may precede a compromise or a critical degradation.
- **Question:** How can a sequence of events likely to lead to a mission failure be identified as early as possible?
- **Objective:** Design a monitoring system correlating several signals: telemetry, network activity, commands, and software behaviour.
- **Possible constraints:** Multiple data sources, noise, and the need to limit false alerts.
- **Evaluation:** Anticipation capability, time gained before a critical incident, alert accuracy, and robustness.

05. CHALLENGE PHASES

The challenge runs over approximately seven weeks from kick-off to the final pitch, with a mandatory evaluation checkpoint at the mid-point.

Phase	Timing	Focus
Phase 1	26 August- 2 september	Registration, team formation, and track selection.
Phase 2	2 September - 26 September	Research, design, and initial development.
Mid-review	26 September - 30 September	Mid-way evaluation and pre-selection (elimination round).
Phase 3	1 October - 14 October	Full development of the prototype, model, or demonstrator.
Phase 4	D-Day, at the congress	Final pitch before the jury.

Phase 1: Registration and Track Selection

- **Objective:** Form teams and commit to one problem statement.
- **Submission requirements:**
 - Team name and the selected track and problem statement.
 - Short statement of why the team chose this problem.
 - Brief description of the intended approach.
 - CV of each team member.
- **Deliverable:** Registration form and initial declaration submitted via the official technical challenge registration form.

05. CHALLENGE PHASES

Phase 2: Research and Initial Development

Objective:

Understand the problem in depth, review existing work, and begin building. During this phase teams are expected to:

- Review the state of the art relevant to their problem statement.
- Define their technical approach and justify the choices made.
- Set up their simulation or experimental environment.
- Begin implementation and produce early results.

Mandatory Deliverables:

1. Pitch Video (Max 2 Minutes):

A concise video presentation explaining the problem statement, the core idea, and the proposed technical approach.

1. Interim Research Paper:

Submissions should follow standard academic paper conventions (IEEE double-column format is recommended but not mandatory). Papers should thoroughly articulate the problem, existing benchmarks, and proposed engineering strategy.

- Title & Authors: Project title, track, and team member details.
- Abstract & Keywords: Concise summary (150–250 words) covering the problem, method, and key findings, with 3–5 keywords.
- I. Introduction: Context, problem statement, and main contributions.
- II. Related Work: Review of existing solutions, key limitations, and your added value.
- III. Methodology: System architecture, algorithms, and design choices.
- IV. Experiments & Roadmap: Testbed/simulation setup, preliminary results (if prototyping has started), and validation plans for Phase 3.
- V. Conclusion: Core insights and next milestones.
- References: Academic and technical citations.

Special Recognition: A dedicated recognition will be presented at the congress for the Best Scientific Paper to honor outstanding technical writing, scientific depth, and methodology.

Mid-Way Evaluation and Pre-Selection

- After three to four weeks from kick-off, every team is evaluated. This checkpoint is an elimination round: **only teams scoring at least 60 out of 100 continue to the final phase.**

Phase 3: Final Development

- **Objective:** Deliver a working prototype, model, or demonstrator over the remaining period from 1st october to 14th.
- Teams execute the plan they defended at the mid-way evaluation, incorporating jury feedback. Deliverables at the end of this phase:
 - Final technical report.
 - Repository with documented source code, model, or simulation environment.
 - Results and evaluation against the performance indicators of the chosen problem.

Phase 4: Final pitch before the jury.

- Finalist teams will pitch their solution live before an expert jury at the congress, demonstrating their prototype, experimental results, and technical mastery. Each team will defend their engineering choices during an interactive Q&A session to compete for the final awards.

CONGRESS POSTER SESSION (OPEN TO ALL TEAMS)

To foster scientific exchange and innovation, an official Research Poster Session will take place during the congress.

Inclusivity: Every participating team is strongly encouraged to design and present a scientific poster of their idea, regardless of whether they qualify as final pitch finalists in the mid-review.

- **Format & Guidelines:** The full guidelines, visual templates, and structural details for the poster will be shared with all teams shortly after registration.

06. EVALUATION

6.1. Scoring Grid

All evaluations are scored out of 100 points using the grid below.

Criterion	Points	What the jury evaluates
Understanding of the problem	15	Does the team genuinely understand the space problem being addressed?
Relevance and originality of the solution	15	Does the solution really answer the need? Does it bring something relevant?
Technical architecture and design	20	Is the system correctly designed and structured?
Feasibility and realism	15	Can it be achieved with the resources and time available?
Actual prototype progress	20	Is there concrete proof of development?
Methodology and planning	10	Does the team clearly know how to reach the final result?
Presentation and command of the project	5	Ability to explain, defend, and justify the choices made.
Total	100	

Passing Threshold

A minimum score of 60 out of 100 is required to pass the pre-selection and continue to the final phase. Teams scoring below this threshold are eliminated at this stage.

07. PARTICIPATION GUIDELINES

Team Composition

- Teams may include up to 5 members.

Submission Guidelines

- Registration. All participants must complete their registration through the official form shared by the organisers. Only registered teams are granted access to the submission forms.
- Deadlines. All submissions must respect the given deadlines. After the deadline, the submission is not accepted and no points are awarded for that phase.

Bonus Points

- Having at least one IEEE IAS YP Member(1pt)

Join us and be part of this quest for excellence to shape a technological and innovative future together.

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08. SUGGESTED RESEARCH & READING LIST

The references below are grouped by track to give every team a credible starting point. Most are recent peer-reviewed articles or arXiv preprints; teams are encouraged to go beyond them.

A. Track 1: AI & Edge / Cloud Computing :

- A Comprehensive Survey on Orbital Edge Computing: Systems, Applications, and Algorithms
<https://arxiv.org/abs/2306.00275>

Keywords: Orbital Edge Computing, LEO Satellites, Computation Offloading, Resource Allocation, Task Scheduling, COTS. Hardware, Satellite Networks

- Trabant: A Serverless Architecture for Multi-Tenant Orbital Edge Computing
<https://arxiv.org/abs/2504.08337>

Keywords: Serverless Computing, Function-as-a-Service, Orbital Edge Computing, Multi-Tenancy, Earth Observation, Time-Shifted Computing, Resource Scheduling

- Satellite Computing: A Case Study of Cloud-Native Satellites
<https://arxiv.org/abs/2307.08530>

Keywords: Cloud-Native Satellites, Satellite Computing, Containerization, Edge Computing, Tiansuan Constellation, On-Orbit Processing, LEO

- Space AI: Leveraging Artificial Intelligence for Space to Improve Life on Earth
<https://arxiv.org/abs/2512.22399>

Keywords: Space AI, Onboard Intelligence, Orbital Data Centers, Autonomous Operations, Machine Learning, In-Space Cloud, Satellite Autonomy

- Advancing Earth Observation: A Survey on AI-Powered Image Processing in Satellites
<https://arxiv.org/abs/2501.12030>

Keywords: Earth Observation, Onboard AI, Edge Computing, Image Processing, Machine Learning, Satellite Remote Sensing

- FPGA-Based Neural Network Accelerators for Space Applications: A Survey
<https://arxiv.org/abs/2504.16173>

Keywords: FPGA, Neural Network Accelerators, Onboard Processing, Radiation Effects, Fault Tolerance, COTS, Edge AI

B. Track 2: Distributed & Federated Machine Learning :

- Graph Theory Meets Federated Learning over Satellite Constellations: Spanning Aggregations, Network Formation, and Performance Optimization
<https://arxiv.org/abs/2509.24932>

Keywords: Federated Learning, Satellite Constellations, Graph Theory, Spanning Tree Aggregation, Network Formation, Convergence Analysis, Latency Optimization

- Satellite Federated Fine-Tuning for Foundation Models in Space Computing Power Networks
<https://arxiv.org/abs/2504.10403>

Keywords: Foundation Models, Federated Fine-Tuning, Model Parallelism, Inter-Satellite Links, Latency Minimization, Space Computing Power Network, Remote Sensing

- Orbit-Aware Split Learning: Optimizing LEO Satellite Networks for Distributed Online Learning
<https://arxiv.org/abs/2501.11410>

Keywords: Split Learning, Distributed Training, LEO Satellites, Energy Efficiency, Model Partitioning, Online Learning, Resource Optimization

08. SUGGESTED RESEARCH & READING LIST

AC. Track 3: Orbital Data-Center Architecture :

- When to Compute in Space

<https://arxiv.org/abs/2512.17054>

Keywords: Orbital Data Centers, Compute Placement, Multi-Criteria Optimization, Edge-Cloud Continuum, Workload Scheduling, Latency, Utility Function

- Which Workloads Belong in Orbit? A Workload-First Framework for Orbital Data Centers Using Semantic Abstraction

<https://arxiv.org/abs/2603.20317>

Keywords: Orbital Data Centers, Workload Suitability, Semantic Abstraction, Space-Based Compute, GPU Workloads, Downlink Reduction, LLM Workloads

- Towards a Future Space-Based, Highly Scalable AI Infrastructure System Design (Project Suncatcher)

<https://arxiv.org/abs/2511.19468>

Keywords: Space-Based AI, TPU, Free-Space Optics, DWDM, Satellite Constellation, Radiation Testing, Inter-Satellite Links

D. Track 4: Networking & Communication :

- Semantic Communication Enabled 6G-NTN Framework: A Novel Denoising and Gateway Hop Integration Mechanism

<https://arxiv.org/abs/2409.14726>

Keywords: Semantic Communication, 6G, Non-Terrestrial Networks, Satellite Communication, Deep Learning, Bandwidth Efficiency, Gateway Hop Relay

- Analysis of Laser Intersatellite Links and Topology Design for Mega-Constellation Networks

<https://ui.adsabs.harvard.edu/abs/2024IITJ...1134554Y/abstract>

Keywords: Laser Inter-Satellite Links, Optical Networking, Mega-Constellation, Topology Design, Free-Space Optics, LEO Networks, Network Capacity

- Integration of Data Center into the Distributed Satellite Cluster Networks: Challenges, Techniques, and Trends

<https://ieeexplore.ieee.org/document/9887918/>

Keywords: Satellite Data Center Networks, Distributed Satellite Clusters, Traffic Grooming, SDN, Wavelength Assignment, Optical Networking

- Free Space Optical Communication for Inter-Satellite Link: Architecture, Potentials and Trends

<https://ieeexplore.ieee.org/document/10462050>

Keywords: Free-Space Optics, Inter-Satellite Links, Laser Communications, Pointing and Tracking, Modulation, 6G

08. SUGGESTED RESEARCH & READING LIST

E. Track 5: Cybersecurity

- Securing Satellite Link Segment: A Secure-by-Component Design (link & signal security)

<https://arxiv.org/abs/2411.12632>

Keywords: Satellite Link Security, Free-Space Optics, Eavesdropping, Jamming, Secure-by-Design, Inter-Satellite Links, Physical Layer Security

- SATversary: Adversarial Attacks and Defenses for Satellite Fingerprinting (AI/ML model security)

<https://arxiv.org/abs/2506.06119>

Keywords: Adversarial Attacks, Machine Learning Security, Satellite Fingerprinting, Physical-Layer Authentication, Deep Learning, Spoofing, Model Robustness

- SpyChain: Multi-Vector Supply Chain Attacks on Small Satellite Systems (hardware & supply-chain security)

<https://arxiv.org/abs/2510.06535>

Keywords: Supply Chain Security, Hardware Security, Small Satellites, Malware, Telemetry Exfiltration, Zero Trust, Anomaly Detection

- Space Fabric: A Satellite-Enhanced Trusted Execution Architecture (confidential computing)

<https://arxiv.org/abs/2603.23745>

Keywords: Confidential Computing, Trusted Execution Environment, Remote Attestation, Orbital Infrastructure, Zero Trust, Multi-Tenant Security, Key Management

- Cybersecurity of Satellite Communications Systems: A Comprehensive Survey of the Space, Ground, and Links Segments (overview)

<https://ieeexplore.ieee.org/document/10546924>

Keywords: Satellite Cybersecurity, Space Segment, Ground Segment, Link Security, Threat Survey, Zero Trust